

Google removes infected apps

Google removed 17 apps from the Google Play Store for being infected by the Joker (Bread) malware. <http://dgit.in/oct20-21>



Smartphones become more pricey

Reportedly, the average selling price of smartphones globally increased by 10% YoY in Q2 2020. <http://dgit.in/oct20-22>

makes AI good conversationalists such as 'speech synthesis' and 'neural belief'.

Also with future advancements to this technology, AI could deliver the text using emotions and accents to make them more human like. Even though machine learning and big data is being used to teach AI as to what to expect next when conversing with humans, the aspect of delivering the words to humans in a human form is of the essence. Why is it so crucial you may ask, it is mainly because of the end user's stipulation who are attracted towards or expect an overall level of assistance from the speed, accuracy and convenience the AI has to offer? Also the huge surge in the usage of AI in our life has also added to the shift towards voice driven applications in our daily lives.

WHAT IS SPEECH SYNTHESIS AND HOW DOES IT AFFECT THE WAY AI COMMUNICATES WITH HUMANS?

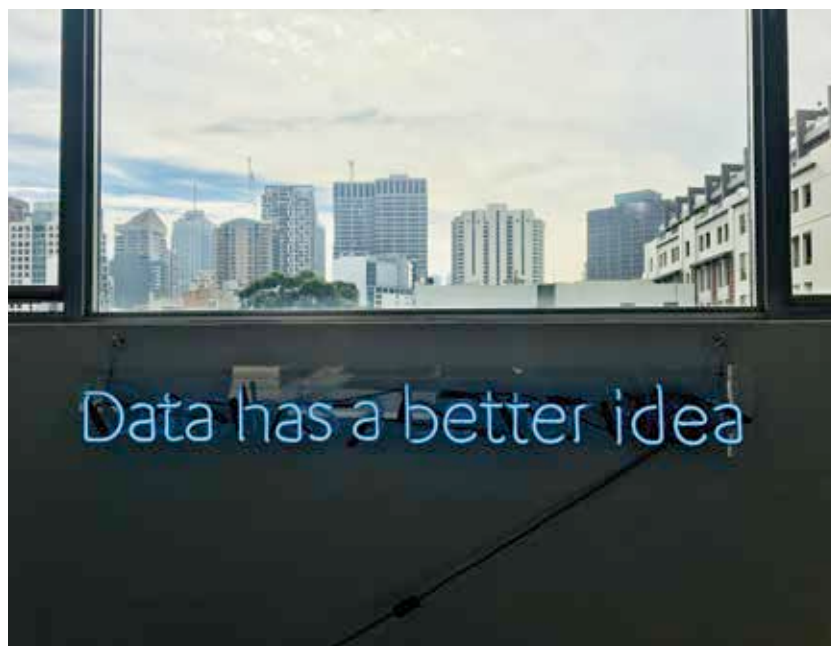
Generally any machine or computer segregates the work they do in three distinct sections which are input (where you upload the information into the computer directly without any further alteration), processing (where the machine or the computer as the name postulates processes the information converting it into a more understandable format) and output (where the computer expresses the input and processed information to the viewer). Speech synthesis just replicates the following processes and is an output method which reads out the words and information processed by the computer. This is more commonly known as a text to speech output. This is very common in today's tech world considering the use of varied machines and computers using TTS to perform simple tasks such as helping us navigate directions and guide blind people. The quality of TTS system synthesizers is determined using four important factors which include the following:

Intelligibility: Is the basically the quality of the audio generated by the computer and also the quality of the punctuation of each words said by the AI. The intelligibility of AI can be evaluated using semantically unpredictable sentences (SUS). They are nothing but passages which consist of unpredictable punctuations and syntaxes to test the reaction of the AI towards such words.

If the SUS are listened to by people they should be able to acknowledge the punctuations and syntaxes present in the speech postulated by the AI. "If semantically predictable sentences are used, listeners will learn the semantic and syntactic cues from the context,

naturalness is a multifaceted, subjective quality of speech, this methodology makes it possible to assess the separate characteristics of the utterance of certain words and sentences.

In two experiments, listeners reliably differentiated the naturalness of speech produced by two male talkers and two text-to-speech systems. Furthermore, they reliably differentiated between the two text-to-speech systems. The conclusion of this experiment suggested that perception of naturalness is only affected by the smallest part of the speech which is essentially the way machines and we humans produce different sounds for each different words produced (glottal pulse).



<https://unsplash.com>

which will influence their performance in the intelligibility task" (Benoît, Grice, & Hazan, 1996).

Naturalness: The timing structure, pronunciation and rendering emotions of the speech given out by the AI is known as the naturalness. Also, the naturalness of the speech will be used in determining the use of the prospective AI. Additionally, the measure and the improvement of synthetic speech is done by running diagnostic tests. Although

Preferences: The preference solely depends on the listeners and how they interpret each TTS with different signal quality and voice. The pattern of results across the four dependent variables – accuracy, search time, naturalness and preference – was consistent and they equally influenced the way how listeners prefer different TTS.

Comprehensibility: The text being read out to the listeners and the listeners actually understanding